

Procedure for ID3 Algorithm

Dataset 1: Student Placement Prediction

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EXPERIMENT 3.py - C:/Users/mohit/OneDrive/Documents/New folder (2)/EXPERIMENT 3.py (3.14.3)
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import pandas as pd
import math

# -----
# Dataset 1: Student Placement Prediction
# -----

data = pd.DataFrame({
    "CGPA": [
        "Good", "Good", "Average", "Poor", "Excellent",
        "Average", "Good", "Excellent", "Poor", "Average"
    ],
    "Communication": [
        "Excellent", "Good", "Good", "Average", "Excellent",
        "Average", "Excellent", "Good", "Poor", "Good"
    ],
    "Programming": [
        "Excellent", "Good", "Good", "Poor", "Excellent",
        "Good", "Good", "Excellent", "Poor", "Average"
    ],
    "Placed": [
        "Yes", "Yes", "Yes", "No", "Yes",
        "Yes", "Yes", "Yes", "No", "No"
    ]
})

# -----
# Entropy Function
# -----

def entropy(data):
    target = data.iloc[:, -1]
    counts = target.value_counts()
    ent = 0

    for count in counts:
        p = count / len(target)
        ent -= p * math.log2(p)

    return ent

# -----
# Information Gain
# -----

def information_gain(data, attribute):
```

```
IDLE Shell 3.14.3
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Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
=== RESTART: C:/Users/mohit/OneDrive/Documents/New folder (2)/EXPERIMENT 3.py ===
Decision Tree:
{'CGPA': {'Good': 'Yes', 'Average': {'Communication': {'Good': {'Programming': {'Good': 'Yes', 'Average': 'No'}}, 'Average': 'No'}}, 'Poor': 'No', 'Excellent': 'Yes'}}

New Sample:
{'CGPA': 'Good', 'Communication': 'Excellent', 'Programming': 'Good'}

Prediction: Yes
>>>
```

Dataset 2: Loan Approval Prediction

```
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import pandas as pd
import math

# -----
# Dataset 2: Loan Approval Prediction
# -----

data = pd.DataFrame({
    "Income": [
        "High", "High", "Medium", "Low", "Medium",
        "High", "Low", "Medium", "High", "Low"
    ],
    "Credit_History": [
        "Good", "Excellent", "Good", "Poor", "Excellent",
        "Good", "Poor", "Good", "Excellent", "Poor"
    ],
    "Employment": [
        "Permanent", "Permanent", "Temporary", "Temporary", "Permanent",
        "Permanent", "Temporary", "Permanent", "Permanent", "Temporary"
    ],
    "Loan Approved": [
        "Yes", "Yes", "Yes", "No", "Yes",
        "Yes", "No", "Yes", "Yes", "No"
    ]
})

# -----
# Entropy Function
# -----

def entropy(data):
    target = data.iloc[:, -1]
    counts = target.value_counts()
    ent = 0

    for count in counts:
        p = count / len(target)
        ent -= p * math.log2(p)

    return ent

# -----
# Information Gain
# -----

def information_gain(data, attribute):
```

```
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>>>
=== RESTART: C:/Users/mohit/OneDrive/Documents/New folder (2)/EXPERIMENT 3 B.py ===
Decision Tree:
{'Income': {'High': 'Yes', 'Medium': 'Yes', 'Low': 'No'}}

New Sample:
{'Income': 'Medium', 'Credit_History': 'Excellent', 'Employment': 'Permanent'}

Loan Approval Prediction: Yes
>>>
```

Dataset 3: Disease Diagnosis

```
EXPERIMENT 3 C.py - C:/Users/mohit/OneDrive/Documents/New folder (2)/EXPERIMENT 3 C.py (3.14.3)
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import pandas as pd
import math

# -----
# Dataset 3 : Disease Diagnosis
# -----

data = pd.DataFrame({
    "Fever": [
        "Yes", "Yes", "No", "Yes", "No",
        "Yes", "No", "Yes", "No", "Yes"
    ],
    "Cough": [
        "Yes", "No", "Yes", "Yes", "No",
        "Yes", "Yes", "No", "No", "Yes"
    ],
    "Body_Pain": [
        "Yes", "Yes", "No", "No", "Yes", "No",
        "Yes", "No", "Yes", "No", "Yes"
    ],
    "Disease": [
        "Positive", "Positive", "Negative", "Positive", "Negative",
        "Positive", "Negative", "Positive", "Negative", "Positive"
    ]
})

# -----
# Entropy Function
# -----

def entropy(data):
    target = data.iloc[:, -1]
    counts = target.value_counts()

    ent = 0

    for count in counts:
        p = count / len(target)
        ent -= p * math.log2(p)

    return ent

# -----
# Information Gain
# -----

def information_gain(data, attribute):
```

```
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>>> == RESTART: C:/Users/mohit/OneDrive/Documents/New folder (2)/EXPERIMENT 3 C.py =
Decision Tree:

{'Fever': {'Yes': 'Positive', 'No': 'Negative'}}

New Sample:
{'Fever': 'Yes', 'Cough': 'Yes', 'Body_Pain': 'Yes'}

Disease Prediction: Positive
>>>
```

Dataset 4: Employee Promotion Prediction

```
import pandas as pd
import math

# -----
# Dataset 4 : Employee Promotion Prediction
# -----

data = pd.DataFrame({
    "Experience": [
        "High", "High", "Medium", "Low", "Medium",
        "High", "Low", "Medium", "High", "Low"
    ],
    "Performance": [
        "Excellent", "Good", "Good", "Average", "Excellent",
        "Excellent", "Average", "Good", "Excellent", "Average"
    ],
    "Training": [
        "Yes", "Yes", "No", "No", "Yes",
        "Yes", "No", "Yes", "Yes", "No"
    ],
    "Promotion": [
        "Yes", "Yes", "Yes", "No", "Yes",
        "Yes", "No", "Yes", "Yes", "No"
    ]
})

# -----
# Entropy Function
# -----

def entropy(data):
    target = data.iloc[:, -1]
    counts = target.value_counts()

    ent = 0

    for count in counts:
        p = count / len(target)
        ent -= p * math.log2(p)

    return ent

# -----
# Information Gain
# -----

def information_gain(data, attribute):
```

```
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>>> == RESTART: C:/Users/mohit/OneDrive/Documents/New folder (2)/EXPERIMENT 3 D.py =
Decision Tree:

{'Experience': {'High': 'Yes', 'Medium': 'Yes', 'Low': 'No'}}

New Employee:
{'Experience': 'Medium', 'Performance': 'Excellent', 'Training': 'Yes'}

Promotion Prediction: Yes
>>>
```

Dataset 5: Email Spam Detection

```
import pandas as pd
import math

# -----
# Dataset 5 : Email Spam Detection
# -----

data = pd.DataFrame({
    "Contains_Link": [
        "Yes", "Yes", "No", "Yes", "No",
        "Yes", "No", "Yes", "No", "No"
    ],
    "Contains_Offer": [
        "Yes", "No", "No", "Yes", "No",
        "Yes", "Yes", "No", "No", "Yes"
    ],
    "Sender_Type": [
        "Unknown", "Known", "Known", "Unknown", "Known",
        "Unknown", "Known", "Unknown", "Known", "Known"
    ],
    "Spam": [
        "Yes", "No", "No", "Yes", "No",
        "Yes", "No", "Yes", "No", "No"
    ]
})

# -----
# Entropy Function
# -----

def entropy(data):
    target = data.iloc[:, -1]
    counts = target.value_counts()

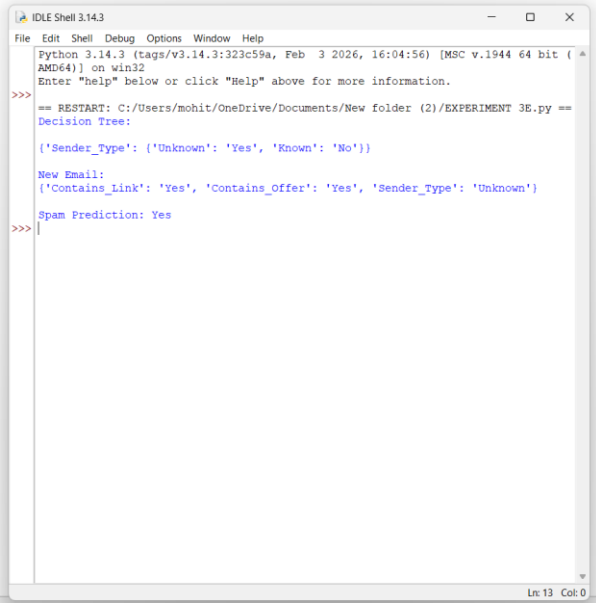
    ent = 0

    for count in counts:
        p = count / len(target)
        ent -= p * math.log2(p)

    return ent

# -----
# Information Gain
# -----

def information_gain(data, attribute):
```



```
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>>> == RESTART: C:/Users/mohit/OneDrive/Documents/New folder (2)/EXPERIMENT 36.py ==
Decision Tree:

{'Sender_Type': ('Unknown': 'Yes', 'Known': 'No')}

New Email:
{'Contains_Link': 'Yes', 'Contains_Offer': 'Yes', 'Sender_Type': 'Unknown'}

Spam Prediction: Yes
>>> |
```